

EMPIRICAL VALIDATION I · 2026

# Pairs Trading, Empirical Validation I

**Six Tests on Real US ETF Data — backtest, allocation, execution, regime, threshold, information share**

Real data on 20 US ETFs closes the limits of Modern Pairs Trading (ten techniques) one by one.

**Empirics**

**Alloc. · Execution**

**Regime · Threshold**

**Information**

Adjusted closes, 2015–2026, 2,949 trading days. Everything not marked otherwise is real data

## EXECUTIVE SUMMARY

# Real data reversed Modern Pairs Trading's central claim

Tracing the cause linked the six tests into one chain.

**What reversed**

The cost-aware optimal band  $0.50\sigma$  lost to the conventional  $\pm 2\sigma$  (Sharpe 0.84 vs 1.11)

**Why**

The reversion rate rises with deviation size, 26%→49%. The spread is not a linear OU

**The formal model**

A threshold ECM. The estimated threshold  $1.88\sigma$  sits where the conventional  $\pm 2\sigma$  does

**What remains**

A Sharpe of 1.11 found by searching 190 pairs has DSR 0.67 — not significant

**Before solving the formula more precisely, test the model assumption it stands on**

## CONTENTS

# Six tests and the limits they close

**S1****Empirical backtest**

Not a simulation

**S2****Capital allocation**

Multiple pairs were left out

**S3****Execution model**

Intraday was out of scope

**S4****Formal regime estimation**

HMM covered only in concept

**S5****Threshold ECM**

Flagged as a next step

**S6****Information share**

Flagged as a next step

# S1

Empirics

## **Real data reveals what the simulation hid**

20 US ETFs · 2,949 trading days · all 190 pairs tested

## S1. Empirics

# Split selection and evaluation by time; freeze the parameters

Shuffling them as in cross-validation leaks future information.

$$\underbrace{[t_0, t_1]}_{\text{select}} \longrightarrow \underbrace{(t_1, t_2]}_{\text{evaluate}}, \quad \hat{\theta} \text{ frozen at } t_1$$

- **Data** Use adjusted closes. Unadjusted prices create fake mean reversion at ex-dividend dates
- **Frozen parameters** Fix hedge ratio, mean, std and band at their selection-period values
- **Pre-registered band** A band chosen after seeing the evaluation period makes its result meaningless
- **Trial log** Count the combinations tested. That number is an input to the DSR
- **Cost sensitivity** Double the costs and see whether the conclusion changes

20 ETFs · 2015–2020 selection (1,511 days) · 2021–2026 evaluation (1,438 days)

## S1. Empirics

# One ADF lag rule doubles the number of pairs that pass

Same data, same test — different result.

| Lag choice            | Pass (of 190) | Share | Verdict                         |
|-----------------------|---------------|-------|---------------------------------|
| <b>AIC, automatic</b> | 10 pairs      | 5.3%  | Equals expected false positives |
| <b>Lag fixed at 1</b> | 25 pairs      | 13.2% | Sees only the fast component    |
| <b>Lag fixed at 5</b> | 17 pairs      | 8.9%  | In between                      |
| <b>Expected at 5%</b> | 9.5 pairs     | 5.0%  | Benchmark under the null        |

- **Implication** The cointegration found among these 20 does not exceed chance

Pick the test specification after the fact and you can get any pass count you like

## S1. Empirics

# Why even an obvious pair fails the test

SPY and IVV track the same index, yet ADF  $p = 0.234$ .

| Lag     | Autocorrelation | Implied half-life | What it sees                         |
|---------|-----------------|-------------------|--------------------------------------|
| 1 day   | 0.939           | 11.0 days         | Fast component — what AR(1) captures |
| 5 days  | 0.932           | 49.0 days         | In between                           |
| 20 days | 0.859           | 91.5 days         | Slow component                       |
| 60 days | 0.696           | 114.6 days        | What the ADF sees                    |

- **Conclusion** The spread is not a single OU but a mix of a fast and a slow component

**No single half-life summarises the series — every later result follows from this**

Overview

**Empirics**

Alloc. · Execution

Regime · Threshold

Information

Synthesis

**S1. Empirics**

# Twenty selected pairs, run at each cost level

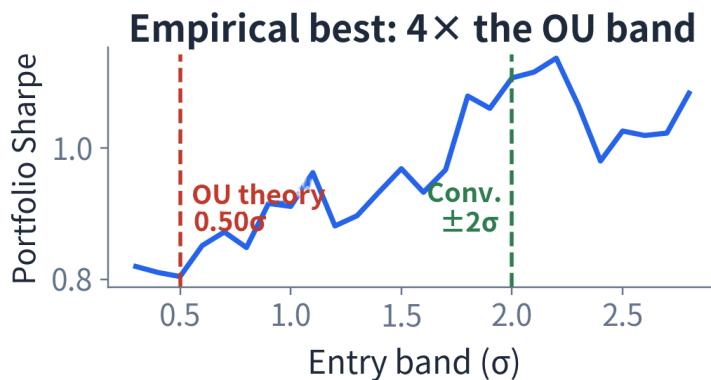
Rolling 90-day z-score with the cost-aware optimal band.

| Round-trip cost | Mean band     | In-sample Sharpe | Out-of-sample Sharpe | Positive | Portfolio Sharpe |
|-----------------|---------------|------------------|----------------------|----------|------------------|
| <b>2bp</b>      | 0.36 $\sigma$ | 0.63             | 0.45                 | 20/20    | 0.95             |
| <b>5bp</b>      | 0.50 $\sigma$ | 0.41             | 0.29                 | 18/20    | 0.84             |
| <b>10bp</b>     | 0.65 $\sigma$ | 0.25             | 0.13                 | 18/20    | 0.66             |
| <b>20bp</b>     | 0.86 $\sigma$ | 0.14             | 0.07                 | 16/20    | 0.44             |

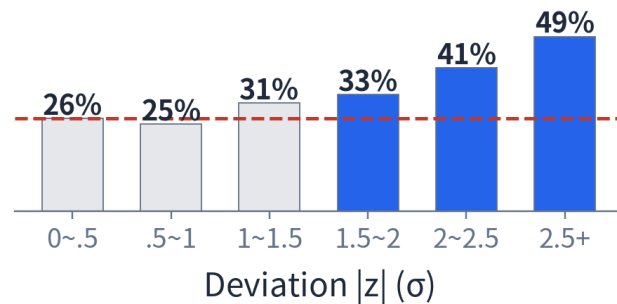
Raising costs from 2bp to 20bp cuts the out-of-sample Sharpe to one sixth

## S1. Empirics

## Modern Pairs Trading's claim ⑦ reverses on real data



**Reversion rises with the deviation**  
Flat if the OU were linear

**0.84**

OU theoretical solution  
 $0.50\sigma$ , 104 trades

**1.11**

Conventional  $\pm 2\sigma$   
39 trades

**2.2 $\sigma$** 

Empirical optimum  
Same at 2, 5, 10bp

**26→49%**

Reversion rate  
Constant if linear

## S1. Empirics

# First we ruled out what was not the cause

Neither an estimation problem nor an objective-function problem.

**1 A  $\kappa$  estimation error?**

Recomputed at lags 1, 5, 20 and 60, the optimal band is  $0.50\sigma$  every time

**2 An objective mismatch?**

Switching from Sharpe to return, the empirical optimum is still  $2.0\sigma$

**3 What remains**

The model itself is wrong — the spread does not follow a linear OU

**4 Check**

Measured by deviation bucket, the reversion rate rises monotonically from 26% to 49%

**Small deviations do not revert; only large ones revert strongly — a threshold structure**

Overview

**Empirics**

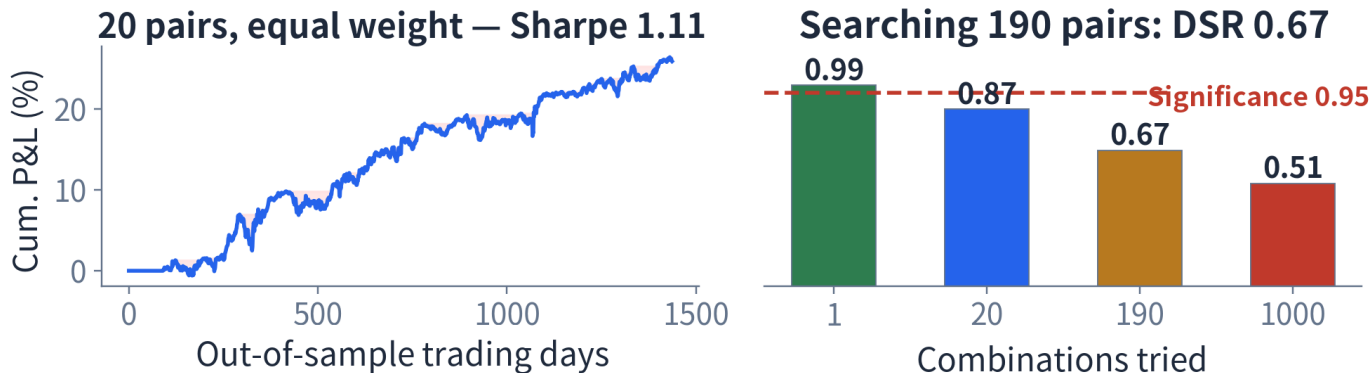
Alloc. · Execution

Regime · Threshold

Information

Synthesis

## S1. Empirics

**Strong point estimate, insignificant after the search cost****1.11**

**Portfolio Sharpe**  
20 pairs, equal weight

**18/20**

**Positive pairs**  
Out-of-sample

**13.4**

**Kurtosis**  
Normal is 3

**0.667**

**DSR (190 trials)**  
Below the 0.95 bar

# S2

Alloc. · Execution

## Many pairs, traded for real

Capital allocation · effective number of bets · execution delay and horizon

## S2. Capital allocation

# The fewer quantities you estimate, the stronger out-of-sample

Equal weight estimates nothing; mean–variance needs expected returns as well.

$$w_i^{\text{EW}} = \frac{1}{N}, \quad w_i^{\text{IV}} = \frac{1/\sigma_i}{\sum_j 1/\sigma_j}, \quad w^{\text{RP}} : w_i(\Sigma w)_i = \frac{1}{N} w^\top \Sigma w$$

$$\text{ENB} = \frac{1}{\sum_k p_k^2}, \quad p_k = \frac{\lambda_k}{\sum_j \lambda_j}, \quad \lambda_k = \text{eig}_k(D_w \Sigma D_w)$$

- **Measure on P&L** Build the covariance from daily strategy P&L, not spread correlations
- **Remove duplicates** Pairs correlated above 0.95 are effectively the same bet
- **Effective bets** Report diversification as ENB, not as the number of pairs

The allocation method itself must be chosen in-sample and evaluated out-of-sample

Overview

Empirics

Alloc. · Execution

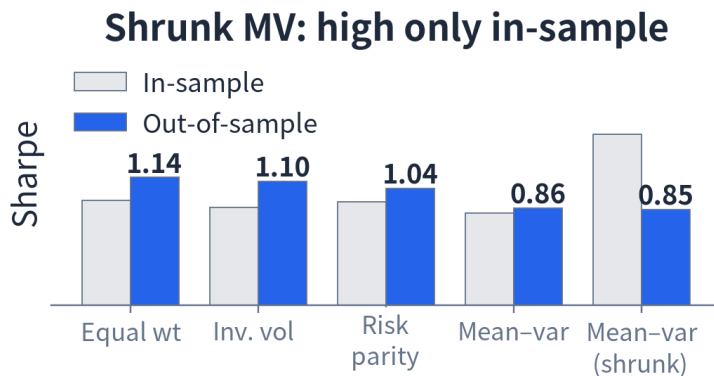
Regime · Threshold

Information

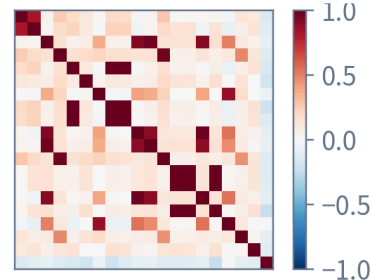
Synthesis

## S2. Capital allocation

## Twenty pairs held, but only 5.3 effective bets



20 pairs = 5.3 effective bets



1.14

Equal weight OS  
Nothing estimated

1.52

Shrunk mean-variance IS  
Best in-sample

0.85

Shrunk mean-variance OS  
Worst out-of-sample

0.998

Max correlation  
SPY·IVV·VOO·HYG

Overview

Empirics

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Synthesis

## S2. Capital allocation

# Best in-sample becomes worst out-of-sample

Max drawdown is in multiples of the daily standard deviation.

| Method                 | IS Sharpe | OS Sharpe | Max weight | Effective bets | OS drawdown |
|------------------------|-----------|-----------|------------|----------------|-------------|
| Equal weight           | 0.93      | 1.14      | 5.0%       | 5.3            | 16.8        |
| Inverse volatility     | 0.87      | 1.10      | 39.6%      | 5.9            | 15.6        |
| Risk parity            | 0.92      | 1.04      | 51.3%      | 7.0            | 14.9        |
| Mean–variance          | 0.82      | 0.86      | 35.2%      | 2.8            | 22.6        |
| Mean–variance (shrunk) | 1.52      | 0.85      | 75.2%      | 2.5            | 14.4        |

Expected–return estimation error exceeds covariance error, so mean–variance collapses

## S3. Execution

# Too fast or too slow both cost — the trough is optimal

Square-root impact trades off against price risk on the unfilled remainder.

$$\mathcal{I}(\tau) = \sigma_d \sqrt{\frac{Q}{\text{ADV} \cdot \tau}}, \quad \mathcal{R}(\tau) = \sigma_d \sqrt{\frac{\tau}{3}}$$

$$\tau^* = \arg \min_{\tau} \left[ \mathcal{I}(\tau) + \lambda \mathcal{R}(\tau) \right]$$

- **Timestamps**      Record signal time and execution time separately
- **Per leg**      The two legs differ in ADV, so their optimal horizons differ
- **Slower leg**      Execute each leg at its own optimum and the hedge breaks in between

Log realised implementation shortfall on every trade and update the backtest cost assumption

Overview

Empirics

Alloc. · Execution

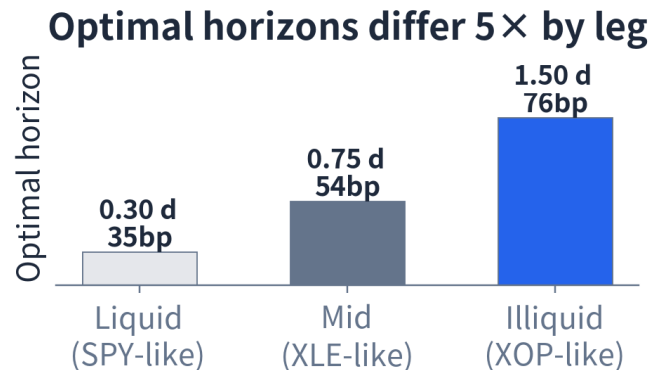
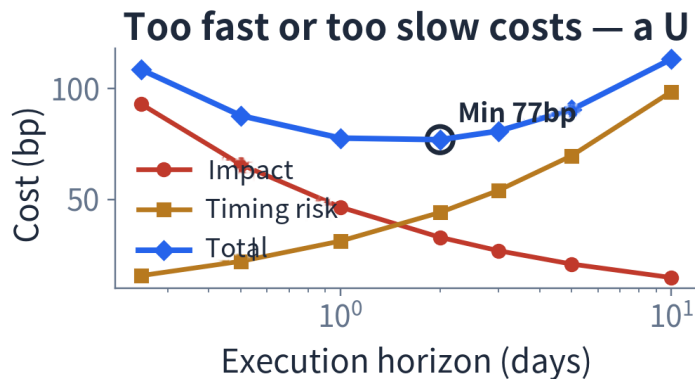
Regime · Threshold

Information

Synthesis

## S3. Execution

## | One day late cuts the Sharpe by 8%



1.14

Same-day fill  
Portfolio Sharpe

1.05

1-day delay  
-8%

0.93

2-day delay  
-18%

5×

Horizon gap  
0.30 vs 1.50 days

# S4

Regime · Threshold

## Estimating the nonlinearity formally

Markov-switching model · threshold ECM · no-arbitrage band

## S4. Regime

# The choice of observed variable matters most

The filtered probability updates one observation at a time.

$$y_t \mid S_t = k \sim \mathcal{N}(\mu_k, \sigma_k^2), \quad \Pr(S_t = j \mid S_{t-1} = i) = p_{ij}$$

$$\xi_{t|t}(k) = \frac{f(y_t | S_t = k) \xi_{t|t-1}(k)}{\sum_j f(y_t | S_t = j) \xi_{t|t-1}(j)}$$

- **States** Start with 2. Three or more are hard to interpret and to converge
- **Start values** Fit many times. A variance collapsing to 0 is a degenerate solution
- **Confirm rule** Trading straight off the raw filtered probability gives many false alarms
- **Joint report** Look at false-alarm rate and detection delay together

Evaluate with the real-time filtered probability — the smoothed probability leaks future information

Overview

Empirics

Alloc. · Execution

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Synthesis

## S4. Regime

# Pick the wrong observed variable and the sign flips

Data from Modern Pairs Trading ©: calm sd 1.0, half-life 4 days → stress sd 2.0, half-life 22 days.

| Observed variable             | Calm  | Stress | Reading                    |
|-------------------------------|-------|--------|----------------------------|
| Steady-state sd (true)        | 1.00  | 2.00   | Doubles                    |
| Rolling 20-day std            | 0.82  | 0.70   | Actually falls             |
| Daily innovation sd           | 0.589 | 0.502  | Shrinks as reversion slows |
| Mean absolute deviation $ x $ | 0.62  | 1.76   | Separates cleanly          |

Watching short-term volatility alone reads the stress regime as calmer

Overview

Empirics

Alloc. · Execution

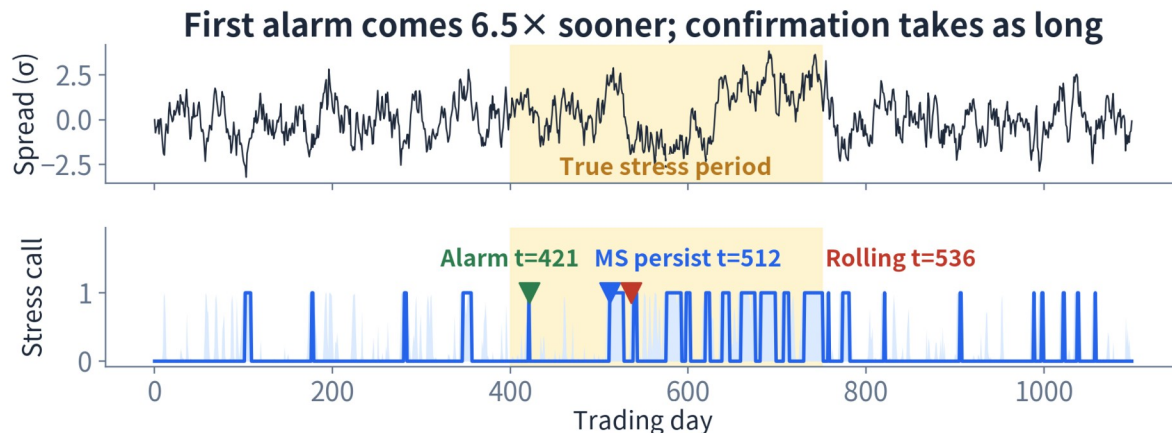
Regime · Threshold

Information

Synthesis

## S4. Regime

## It gives early warning, not early confirmation

**136 days**Rolling half-life  
Accuracy 72.6%**21 days**MS + 5-day confirm  
Accuracy 74.5%**112 days**MS persistence rule  
Close to rolling**6%**False-alarm rate  
Rolling 7%

## S5. Threshold ECM

# No adjustment happens inside the band

In theory the threshold is a no-arbitrage band.

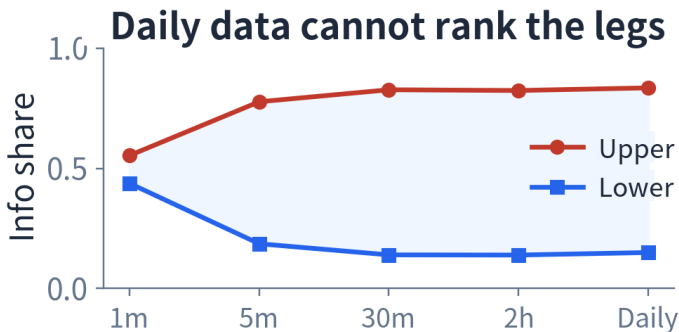
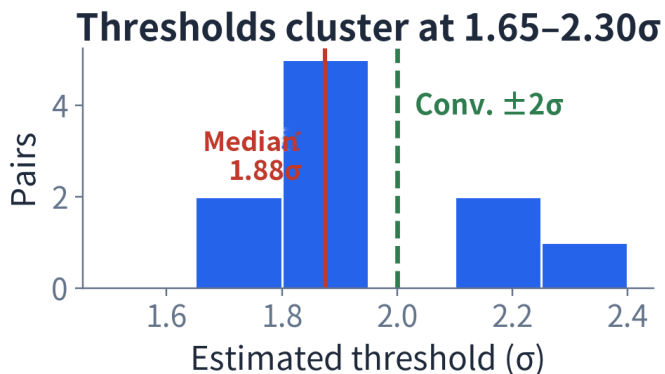
$$\Delta s_t = \begin{cases} \alpha_{\text{out}} z_{t-1} + \varepsilon_t & |z_{t-1}| > \theta \\ \alpha_{\text{in}} z_{t-1} + \varepsilon_t & |z_{t-1}| \leq \theta \end{cases}$$

- **Coordinates** Build it the same way as the trading  $z$  (rolling mean)
- **Min. sample** Each regime needs 8–10% of observations for its coefficient to be identified
- **Linear test** If the likelihood ratio is not significant, there is no case for a threshold
- **Signs** Outside: negative and significant; inside: indistinguishable from 0
- **Band fit** Entering inside the threshold bets on a zone with no adjustment

Inside the band, trading costs keep arbitrageurs from acting

## S5. Threshold ECM

The estimated threshold sits where the conventional  $\pm 2\sigma$  does

 **$1.88\sigma$** 

Median threshold  
Range  $1.65\text{--}2.30$

 **$2.0\text{--}2.2\sigma$** 

S1 empirical optimum  
Same place

**4/10**

Sign pattern matches  
Limit of daily data

**13×**

Adjustment ratio, those 4  
Outside vs inside

## S5. Threshold ECM

# Convention was closer to the true model than theory

The formula in Modern Pairs Trading ⑦ was not wrong. The model it assumed was.

## What a linear OU assumes

- Reversion speed is proportional to the deviation
- Small deviations revert at the same rate
- → optimal band  $0.50\sigma$
- → empirical portfolio Sharpe 0.84

## What the threshold model says

- No adjustment inside the band
- Strong reversion only outside it
- → threshold  $1.88\sigma$
- → empirical portfolio Sharpe 1.11

## Practical implication

Ten lines of code measure the reversion rate by bucket, and those ten lines made a bigger difference than the whole band optimisation

# S6

Information

## Which leg carries the signal

Permanent–transitory decomposition · information share · the frequency problem

## S6. Information share

# The leg that doesn't adjust holds the information

The leg with the smaller absolute adjustment coefficient leads price discovery.

$$\Delta y_{1t} = \alpha_1 z_{t-1} + \dots, \quad \Delta y_{2t} = \alpha_2 z_{t-1} + \dots, \quad \gamma = \frac{(\alpha_2, -\alpha_1)}{\alpha_2 - \alpha_1}$$

$$IS_1 = \frac{([\gamma F]_1)^2}{\gamma \Omega \gamma^\top}, \quad \Omega = FF^\top, \quad F \text{ Cholesky}$$

- **Enough lags** Too few leave correlation in the residuals and distort the information share
- **Signs first** If the two adjustment coefficients share a sign, the decomposition fails
- **Both bounds** Reporting one presents the outcome of an ordering choice as fact

The leg that follows price discovery is the one that mean-reverts — the signal comes from it

## S6. Information share

## Daily data cannot rank the legs

Contemporaneous residual correlation across the 10 empirical pairs is high, 0.63–0.77.

| Item                                  | Value      | Meaning                                                        |
|---------------------------------------|------------|----------------------------------------------------------------|
| <b>GG weight inside [0,1]</b>         | 5/10 pairs | In the rest the coefficients share a sign, so no decomposition |
| <b>Hasbrouck bound width</b>          | Mean 0.51  | SPY–HYG spans 0.24–0.97 — nothing can be said                  |
| <b>Residual contemporaneous corr.</b> | 0.63~0.77  | The higher it is, the wider the bounds                         |
| <b>1-minute (sim)</b>                 | Width 0.12 | Higher frequency identifies it                                 |
| <b>Daily (sim)</b>                    | Width 0.69 | Contemporaneous correlation grows, bounds widen                |

Information share is at heart an intraday-data problem — back to S3

# S1-6

Synthesis

## **One chain made of six tests**

What was confirmed, and what remains

## Synthesis

# The six are not separate topics but one chain

**1** S1 → S5

The band reversed on real data, and its cause, nonlinearity, was formalised as a threshold model

**2** S5 → S4

Regime switching is the same kind of nonstationarity. It warns early but cannot confirm

**3** S1 → S2

Twenty pairs are only 5.3 effective bets. The allocation that estimates least wins

**4** S2 → S3

A day's delay costs 8% of Sharpe. The two legs' optimal horizons differ fivefold

**5** S6 → S3

Which leg carries the signal cannot be identified without intraday data

**Modern Pairs Trading holds, but test its model before optimising the formula**

## Synthesis

# What these six tests still leave open

## Data limits

- Twenty ETFs: a small, overlapping universe
- No intraday quotes: literature impact value
- Two regime switches only: wide intervals
- S3 horizons, S6 frequencies are simulated

## Next

- Retest on single stocks, beyond 190 pairs
- Intraday re-estimate: threshold, info share
- Pre-register a threshold band, walk-forward
- Repeat on Korean stocks and cash-futures

Writing down what stays open becomes the next module's table of contents

Overview

Empirics

Alloc. · Execution

Regime · Threshold

Information

**Synthesis**

## Synthesis

# Settings needed to reproduce

| Item                | Setting                                                                         |
|---------------------|---------------------------------------------------------------------------------|
| <b>Data</b>         | Yahoo Finance daily adjusted closes, 20 ETFs, 2015-01-02 to 2026-09-24          |
| <b>Tickers</b>      | SPY IVV VOO QQQ XLK XLE XOP USO GLD GDX SLV XLF KRE IYR VNQ HYG JNK EEM VWO EWA |
| <b>Periods</b>      | Selection 2015–2020 (1,511 days) / evaluation 2021–2026 (1,438 days)            |
| <b>Costs</b>        | Round trip 2, 5, 10, 20bp (default 5bp), on gross notional                      |
| <b>Trading rule</b> | Rolling 90-day z-score; enter when $ z  > b$ , exit when z changes sign         |

Only the S3 execution horizon and S6 frequency experiments are simulated; everything else is real data

# Test the assumption before solving the formula

Ten lines that measure the reversion rate by bucket mattered more than the whole band optimisation